Job Safety Training Outline

Built to align with DAFI 91-202 paragraph 14.1 requirements.

Unit: 436 CES

Work Center: Engineering

Work Area / Office Symbol: CENME

Phone: 302-677-6828

Supervisor: SrA John N. Hidalgo

Reviewer: SSgt Christopher G. Coney

Review Date: 2026-11-09

Effective Date: 2026-11-09

Work Center Overview

The CES/CENME (Execution Support and GeoBase) work center is a specialized element within the Engineering Flight (CEN) of a Civil Engineer Squadron, largely manned by 3E5X1 Engineering specialists and civilian equivalents . Its primary mission is to bridge technical project design, spatial asset management, and construction project closeout. Personnel in this section direct and execute engineering drafting (CAD/BIM), precise construction and topographic surveying, materials testing (soil, asphalt, and concrete), and contract construction surveillance . Furthermore, CENME serves as the gatekeeper for the project closeout lifecycle—receiving, reviewing, and archiving electronic "as-built" drawings, red-line contractor submittals, and DD Form 1354 real property transfer documentation to ensure every facility modification is permanently documented.

In tandem with project execution support, CENME operates and maintains the installation's GeoBase / Geographic Information System (GIS) architecture. Specialists collect field survey data using robotic total stations and GPS equipment, translating physical infrastructure into Spatial Data Standards for Facilities, Infrastructure, and Environment (SDSFIE) compliant datasets. By maintaining the Common Installation Picture (CIP), CENME tracks critical planimetric assets (airfields, roads, and building footprints)

alongside underground and above-ground utility networks (water, gas, electrical, and storm systems) . This accurate spatial and record drawing repository directly enables safe excavation, emergency response mapping, and master planning across the entire installation.

Work Area Hazard Analysis / JHA Notes

Major Hazards & Findings
Field survey operations present substantial struck-by hazards from vehicular traffic and taxiing aircraft on active flightlines, as well as biological risks including tick bites, insect stings, venomous snakes, and sharp vegetation when surveying in thick brush, high grass, or swamps. Active construction sites expose inspectors to slip, trip, and fall hazards, open trenches, moving heavy equipment, and overhead electrical power lines. In the lab, personnel encounter airborne nuisance dust and crystalline silica during aggregate sieving and sample preparation, along with high ambient noise when operating vibrating shakers or working in industrial flightline zones.

Bioenvironmental Engineering (BE) Survey Notes
BE industrial hygiene surveys for 3E5X1 operations primarily evaluate ambient noise exposure during mechanical soil preparation, sieve shaking, and airfield operations, as well as airborne respirable dust and crystalline silica concentrations during soil processing. BE periodic monitoring verifies that shop dust exposure remains below regulatory limits and evaluates whether administrative rotations, dust collection, or particulate respirators are required during prolonged aggregate sieving.

Controls & Personal Protective Equipment (PPE)
Hazards are managed through strict job safety controls and personal protective equipment. Field personnel must wear ANSI Class 2/3 high-visibility vests, ASTM F2413 safety-toe boots, eye protection, and long pants/sleeves with insect repellent when working in dense vegetation to protect against ticks and thorns. Work in dusty environments requires proper ventilation, dust extraction, or NIOSH-approved N95 particulate respirators, and hearing protection must be worn during noisy laboratory or flightline tasks.

Special Restrictions & Prohibitions
Personnel must possess a valid Airfield Driver’s License (AF Form 483) and maintain two-way radio communication with the Air Traffic Control Tower before entering the airfield or Controlled Movement Area (CMA). All outdoor surveying must cease immediately upon notification of lightning within 5 nautical miles. Personnel must identify and maintain a safe clearance distance from overhead high-voltage power lines when setting up GPS survey rods, leveling rods, and tripods. Never enter an unshored trench exceeding 5 feet () in depth or enter any permit-required confined space without proper certification and an approved permit.

Required Training Topics

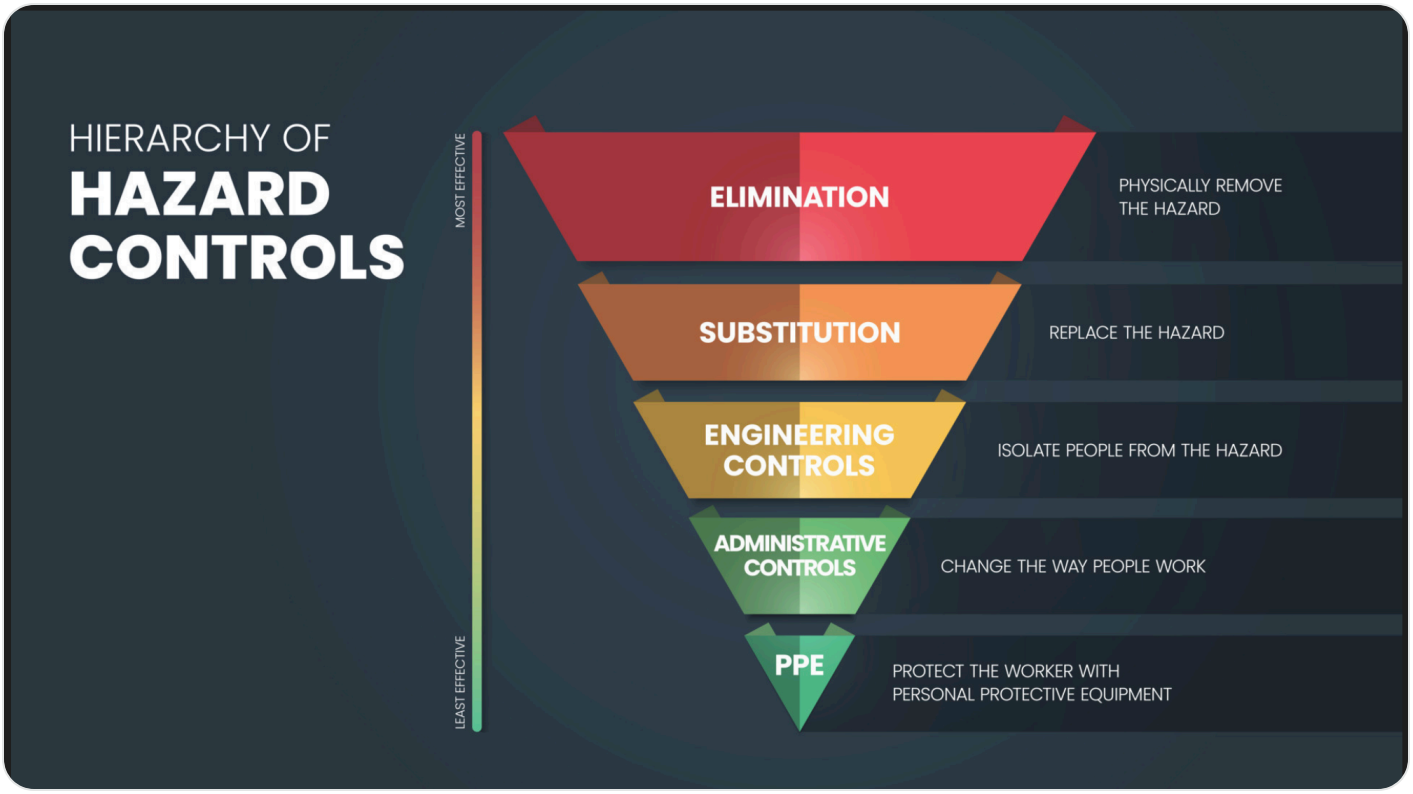
Workplace HazardsPersonal Protective EquipmentEmergency Action and Fire PreventionReporting Unsafe Equipment, Conditions, or ProceduresMishap, Occupational Injury, and Illness ReportingCA-10 and LS-201DAF Traffic Safety ProgramDAFVA 91-209 Location and ContentDAFSMS Responsibilities

- **Workplace hazards:** Describe crushing, burn, chemical, ladder, weather, flight line, and noise hazards specific to the work center, plus the technical orders, JHAs, and instructions workers must follow.
- **Personal protective equipment:** Train employees on required PPE, including donning, doffing, cleaning, maintenance, storage, disposal, and reporting contact-lens or medical issues that could affect wear.
- **Emergency action and fire prevention:** Cover alarms, AEDs, extinguishers, evacuation routes, shelter locations, and response procedures tied to the work center.

- **Reporting unsafe conditions:** Instruct personnel to report unsafe equipment, conditions, procedures, and injuries without fear of reprisal, including use of DAF Form 457 and applicable reporting channels.
- **Mishap / injury / illness reporting:** Explain on-duty and off-duty reporting requirements, medical treatment procedures, and emergency response contacts.
- **Traffic safety:** Brief seat belt, speed limit, spotter, electronic device, and motorcycle safety requirements applicable to the work center.
- **DAFVA 91-209 and DAFSMS:** Review the location of DAFVA 91-209 and how workers contribute to the unit safety management system.

Hierarchy of Controls

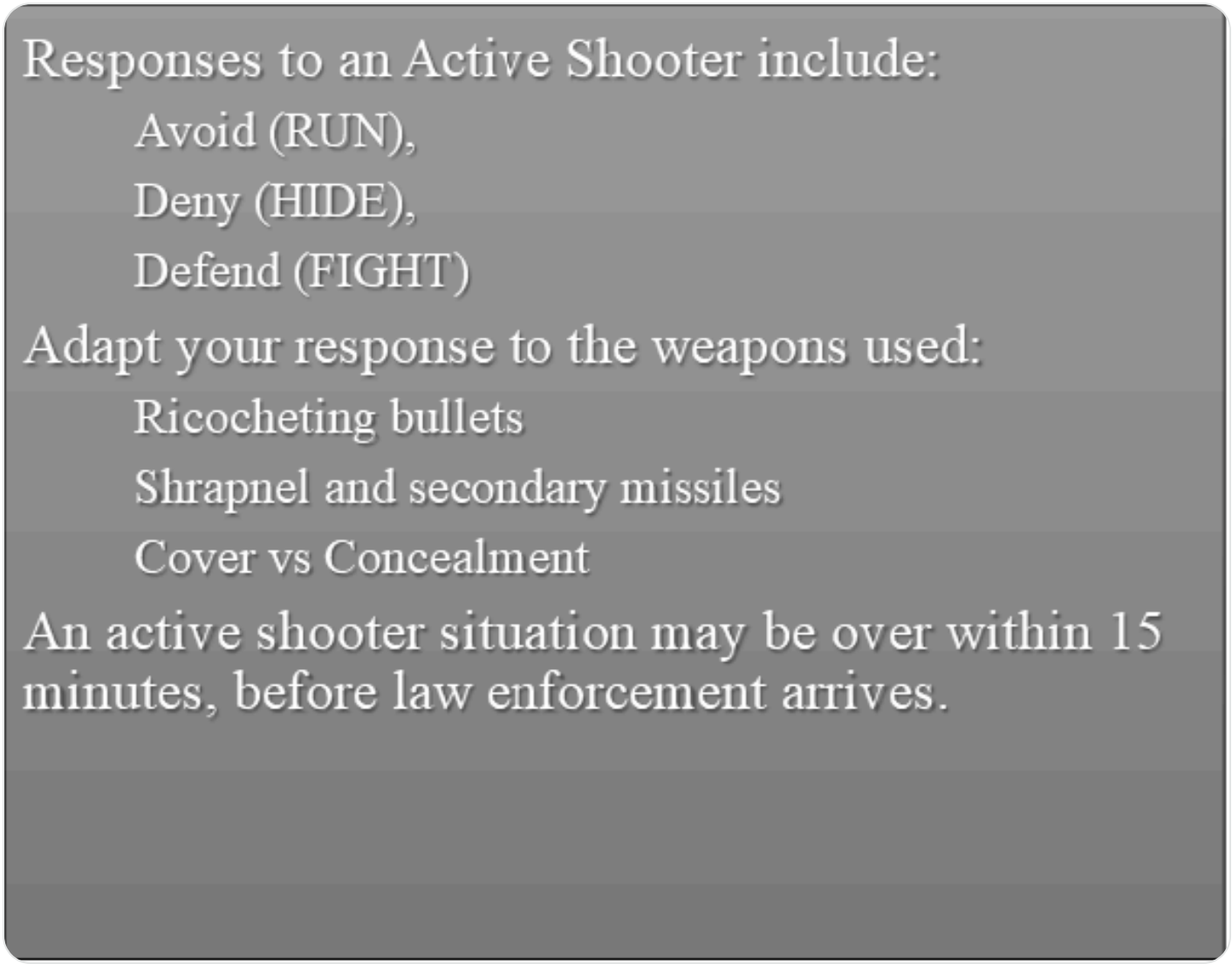
Apply these controls from most effective to least effective when reducing workplace hazards under DAFI 91-202 paragraph 14.1.2.1.4.



Use the highest level of control feasible first. PPE should protect workers only after elimination, substitution, engineering, and administrative controls have been considered.

Active Shooter Response

In the event of an active shooter, individuals should employ the **Avoid, Deny, Defend** response method. This strategy is adaptable to the specific situation and environment.



Avoid (Run): The first and preferred course of action is to evacuate the area if a safe route is available. Maintain awareness of your surroundings and have an escape route planned. Leave your belongings, evacuate regardless of whether others follow, and do not attempt to move wounded people. Once you have reached a safe location, call 911 and provide any information you have.

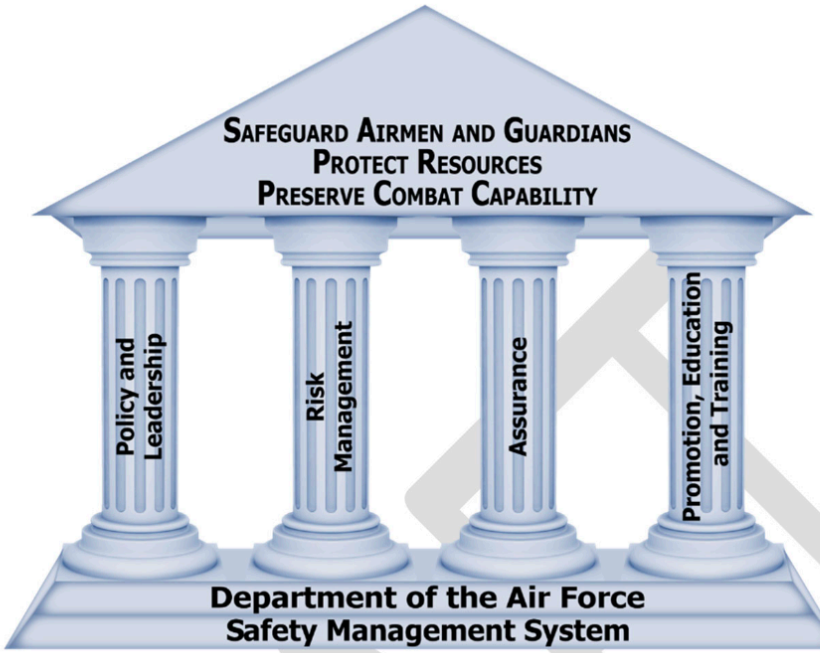
Deny (Hide): If you cannot evacuate, your next priority is to deny the attacker access to your location. This involves more than just hiding; it means securing your position. Lock and blockade doors with heavy furniture, turn off lights, silence your phone, and remain out of the shooter's view. Remember that concealment only hides you, while cover can offer protection from bullets.

Defend (Fight): As a last resort, when your life is in immediate danger, you must be prepared to defend yourself. Act aggressively, improvise weapons from your surroundings, and commit fully to your actions. Taking action may be risky, but it could be your best or only chance for survival.

When law enforcement arrives: Remain calm, follow all instructions, keep your hands visible with fingers spread, and be prepared to provide information about the shooter’s location, number, and description.

DAFSMS Framework

DAFSMS uses four pillars and the DAFSMS framework to structure the mishap prevention program through continuous improvement and the Plan-Do-Check-Act model.



Policy and Leadership

Safety policy provides the structure for a proactive mishap prevention program. Active leadership involvement at every level is critical to implementation and sustained execution.

Risk Management

Integrate risk management into all safety-related activities so hazards are identified early, controls are implemented, and the safety culture stays proactive.

Assurance

Use evaluations, monitoring, and review processes to measure conformance to standards and drive continuous improvement across the mishap prevention program.

Promotion, Education, and Training

Provide safety awareness, embed ongoing training, implement effective controls, and keep airmen and guardians actively engaged in mishap prevention.

Department of the Air Force SMS Framework

Policy and Leadership

- Leadership planning, organization, direction, and control

- Management roles, responsibilities, and relationships
- Procedures and controls that are developed, documented, maintained, and monitored
- Safety and quality balance, integration, and goal setting

Risk Management

- Hazard identification through mission/task analysis and hazard lists
- Hazard assessment using exposure, probability, severity, and risk level
- Control options, effects, prioritization, selection, and decisions
- Implementation, communication, accountability, support, supervision, and feedback

Assurance

- Performance requirements, expectations, and control effectiveness
- Continuous monitoring, self-inspections, internal and external inspections
- Mishap and event investigation, reporting, metrics, and improvement opportunities
- Change management and continuous improvement actions

Promotion, Education, and Training

- Culture built on informed, flexible, learning, just, and reporting behaviors
- Competency through training and education
- Awareness through internal communication, external communication, and information access

Job Specific Training Items

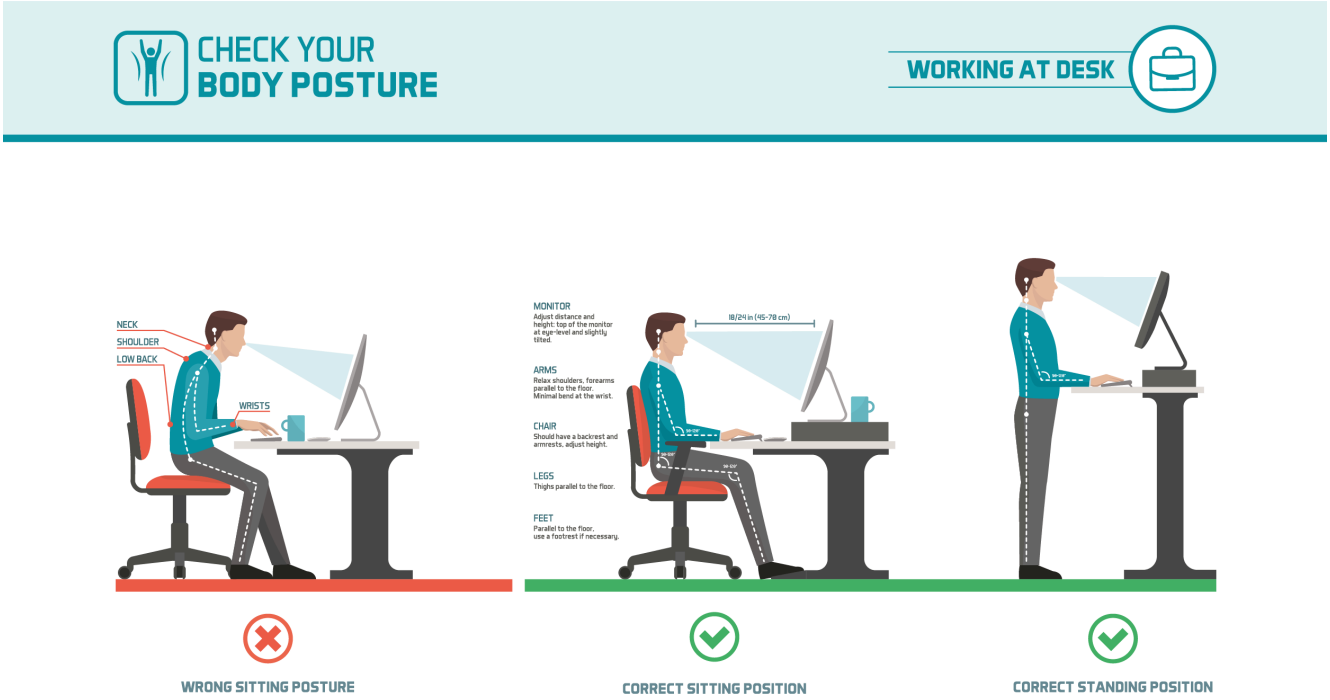
The following modules apply to this work center and should be reviewed with trainees before task performance.

CPR TrainingActive Shooter ResponseGovernment Vehicle / Utility Vehicle UseRisk Management FundamentalsManual Lifting and Material HandlingFire Protection and PreventionPersonal Protective Equipment (PPE)

- **CPR Training:** Ensure designated responders maintain current certification and understand location and use of local AED assets under the unit mishap prevention program. Provide CPR and AED training before assignment where required, and refresh qualification before expiration. DAFMAN 91-203 - Air Force Occupational Safety, Fire and Health Standards, Chapter 1
- **Active Shooter Response:** Use local emergency response procedures, shelter locations, and mass notification methods applicable to the duty location. Brief personnel on how to recognize an active shooter event, how to summon help, and how to apply local run-hide-fight guidance based on the facility layout. Installation emergency management guidance / local procedures
- **Government Vehicle / Utility Vehicle Use:** Brief operators on unit-specific government vehicle training requirements, local route restrictions, low-speed vehicle rules, and documentation of qualification. Train vehicle operators on seat belts, speed limits, local traffic hazards, spotter use while backing, and restrictions on electronic device use while driving. DAFI 91-207; local procedures

- **Risk Management Fundamentals:** Use the DAF risk management process to discuss local hazard reporting, supervisor involvement, and integration into routine tasks. Train workers to identify hazards, assess risk, implement controls, supervise the task, and re-evaluate controls during and after operations. DAFI 90-802; DAFPAM 90-803
- **Manual Lifting and Material Handling:** Provide verbal and written guidance on manual lifting, lifting devices, and local reporting actions for strain or overexertion symptoms. Train workers on proper lifting techniques, team lifting, load assessment, and use of lifting aids to reduce musculoskeletal injury risk. DAFMAN 91-203 - Air Force Occupational Safety, Fire and Health Standards, Chapter 20
- **Fire Protection and Prevention:** Use the JSTO and local fire prevention plan to brief workers on work-center fire hazards, alarm reporting, extinguisher use, and evacuation accountability. Train personnel on fire hazards, reporting procedures, extinguisher locations, evacuation routes, pull stations, and local emergency response actions. DAFMAN 91-203 - Air Force Occupational Safety, Fire and Health Standards; local fire prevention plan
- **Personal Protective Equipment (PPE):** Use the work-center JHA, BE occupational health assessment, manufacturer instructions, DAFMAN 91-203 Chapter 14, and local safety office guidance to identify PPE requirements. Supervisors must ensure PPE is provided, used, cleaned, inspected, maintained in sanitary and serviceable condition, and documented in JSTO training records. Cover eye and face protection, head protection, foot protection, hand protection, protective clothing, and respiratory or hearing protection when those programs apply. Train each affected worker on when PPE is necessary, what PPE is necessary, how to properly don, doff, adjust, and wear PPE, the limitations of assigned PPE, and the proper care, maintenance, useful life, and disposal of PPE. Require workers to demonstrate understanding and proper use before performing work requiring PPE, and retrain when workplace conditions, job assignments, PPE types, or worker knowledge gaps make prior training obsolete. DAFMAN 91-203 - Air Force Occupational Safety, Fire and Health Standards, Chapter 14; 29 CFR Part 1910 - Occupational Safety and Health Standards (General Industry Standards) Subpart I; 29 CFR 1910.132; 29 CFR 1910.133; 29 CFR 1910.135; 29 CFR 1910.136; 29 CFR 1910.138

Ergonomics Workstation Posture Guide:



Use this guide while briefing office and administrative workstation setup.

References

Career Field & Training
• CFETP 3E5X1
• AFQTP 3E5X1 Modules
• DAFI 36-2670

Civil Engineer & Project Execution
• DAFI 32-1023
• DAFI 32-1020
• DAFI 32-1015
• DAFI 32-9005
• DAFI 10-210

GeoBase, CAD/BIM & Design Standards
• DAFI 32-10112 (or current GeoBase Guidance)
• SDSFIE & A/E/C CAD Standards
• UFC 1-200-01 & UFC 3-260-01

Safety, Radiation & JHAs (OSH)
• DAFMAN 91-203 - Air Force Occupational Safety, Fire and Health Standards
• DAFI 91-202 - The Department of the Air Force (DAF) Mishap Prevention Program
• AFMAN 48-148
• AFI 48-139 - Laser and Optical Radiation Protection Program

Technical Orders (TOs), ASTM & OIs
• TO 00-5-1 & TO 00-20-14
• Commercial Manuals / ASTM Standards
• Local Base OIs

Required Documents

- CA-10
  What a Federal Employee Should Do When Injured at Work. Use this as the workplace reference for employee injury reporting and follow-up actions.
- DAF Form 457
  Hazard Report. Use this form to identify, document, and elevate unsafe conditions, equipment, or procedures in the workplace.
- DAFVA 91-209 **UNITS MUST FILL OUT THEIR OWN LOCATION POSTED**
  Department of the Air Force Occupational Safety and Health Program visual aid. Keep this available so personnel know the core safety rights, responsibilities, and reporting expectations.
  Location Posted: Posted on board on hallway 134.
- LS-201
  Notice of Employee's Injury or Death (Non-Appropriated Funds). Use this when applicable for NAF employee injury or death reporting.

Emergency Information

- **Emergency Numbers:** Emergency Response: 911
  BDOC: 302-677-6664
- **Medical Facility / Treatment:** Urgent Care
  1. Bayhealth Walk-in, Blue Hen
  665 S Bay Rd, Unit 3
  Dover, DE 19901

  2. Christiana Care-GoHealth Urgent Care
  15 S Dupont Hwy
  Dover, DE 19901

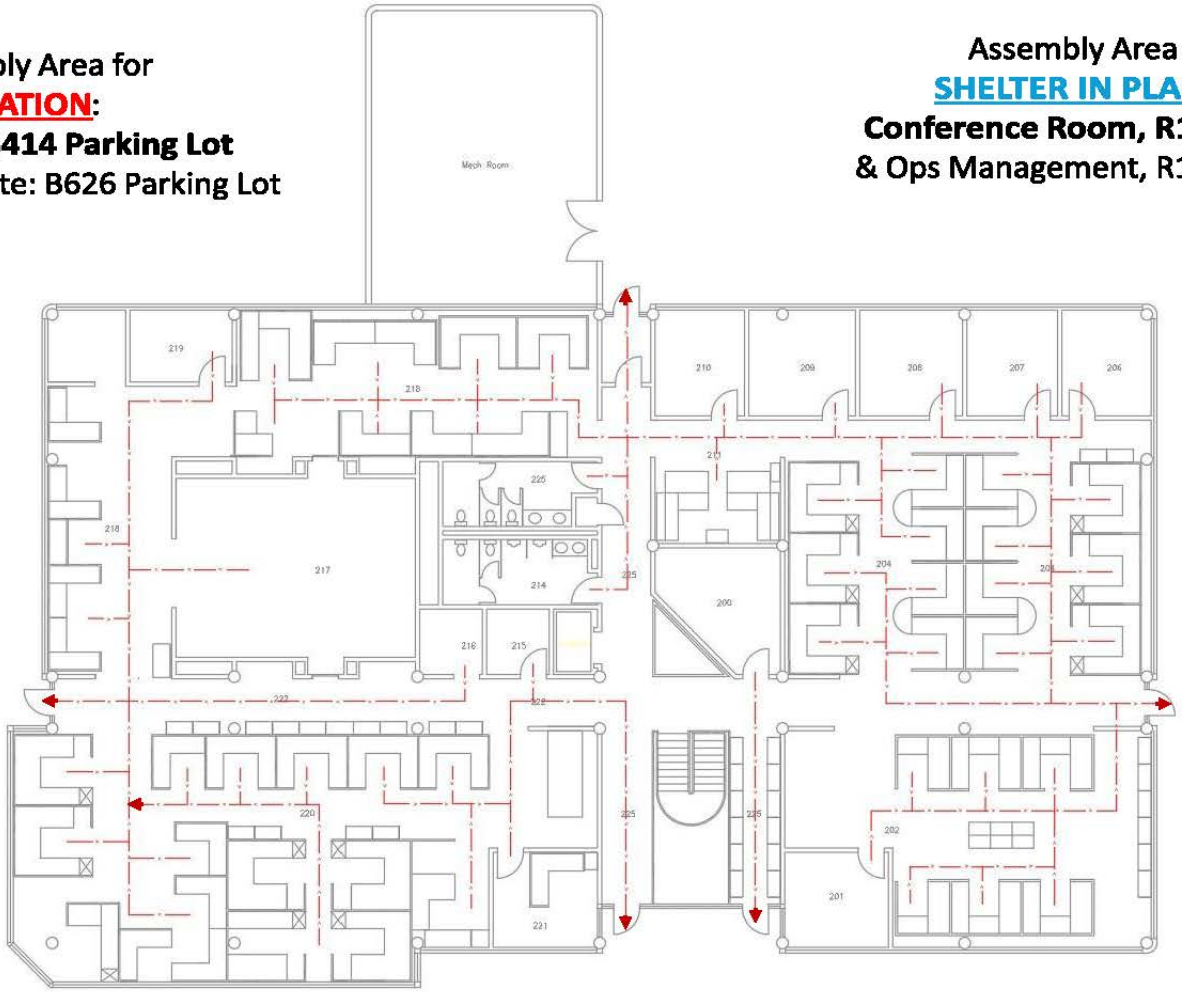
  Emergency Room
  Bayhealth Emergency and Trauma Center, Kent Campus
  625 S Governors Ave
  Dover, DE 19901
- **Evacuation / Muster Point:** Main Evacuation Point: B414 Parking Lot
  Alternate Evacuation Point: B626 Parking Lot
- **Shelter in Place:** Shelter-In-Place Point: Conference Room, R118 & Ops Management, R132
- **Active Shooter Response Methods:** Not entered
- **Adverse Weather Shelter:** Seek shelter in SIP point (Conference Room, R118)

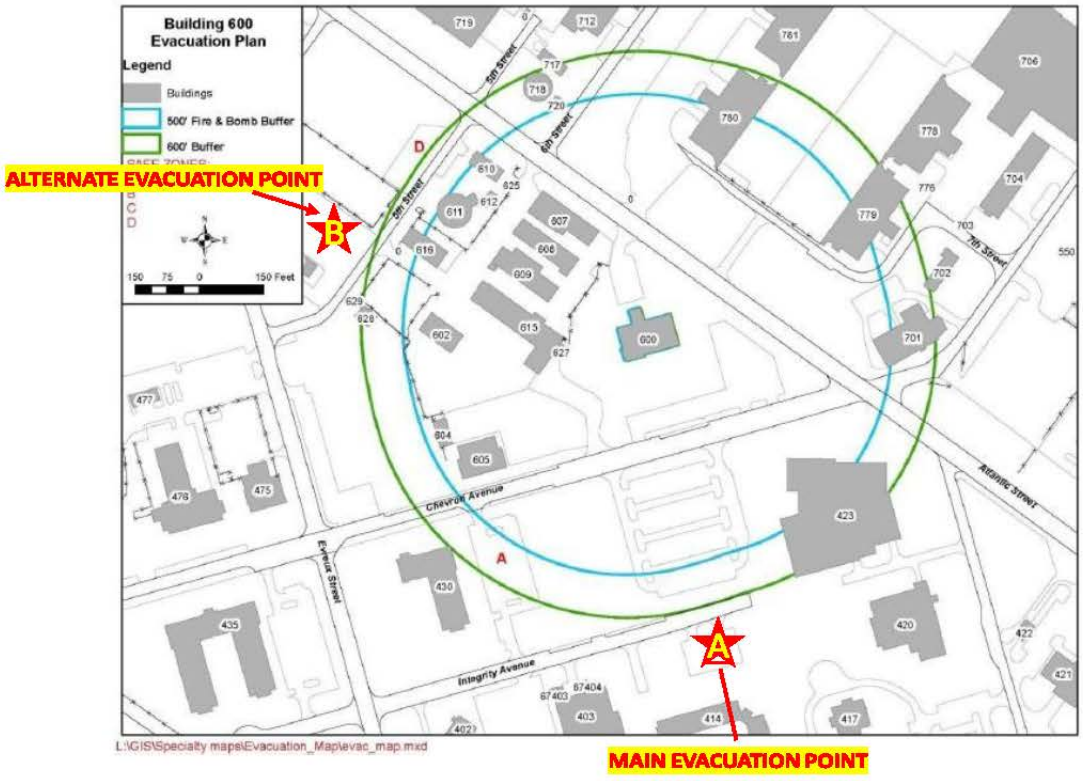
Emergency Evacuation Route Image:

SECOND FLOOR – BUILDING 600

Assembly Area for
EVACUATION:
Main: **B414 Parking Lot**
Alternate: B626 Parking Lot

Assembly Area for
SHELTER IN PLACE:
Conference Room, R118
& Ops Management, R132





Evacuation Plan_2026_Page_1.jpg

Emergency Equipment and Safety Board

Safety Bulletin Board Location: Not entered

List fire pull stations, extinguishers, power cutoffs, eyewash stations, AEDs, foam systems, and any local fire/emergency equipment notes.

No emergency equipment maps or location photos uploaded.

Traffic Safety Program

DAF Traffic Safety Program (DAFI 91-207): Train personnel on mandatory use of seat belts and helmets, speed limits, local traffic hazards, spotters while backing, vehicle training requirements, and restrictions on electronic device use while operating vehicles on- or off-base.

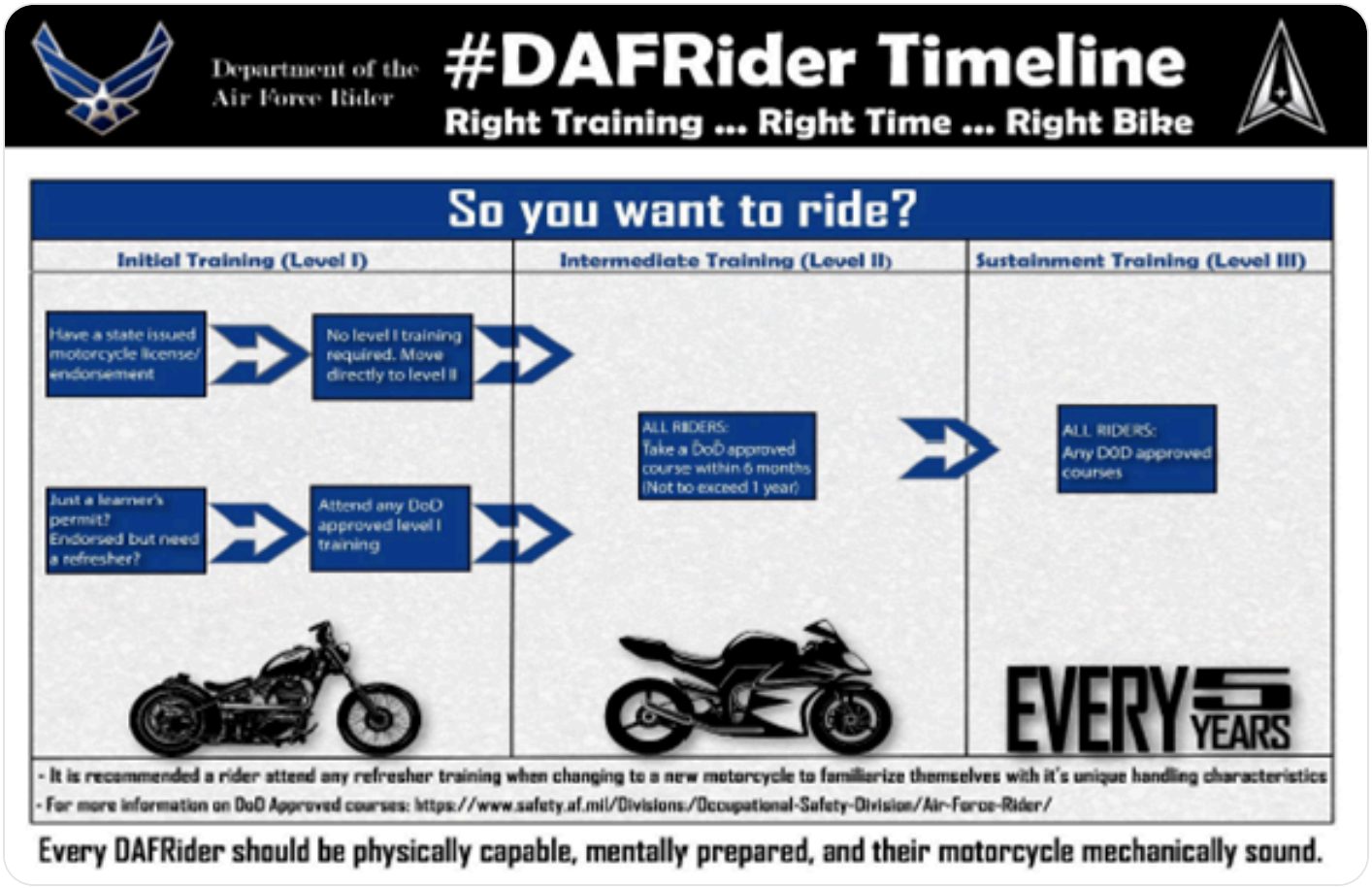
Seat Belts / Occupant Restraints: All persons operating or riding in motor vehicles must wear installed restraint systems as prescribed by the manufacturer. Drivers are responsible for ensuring all passengers are briefed on and use required restraints.

Motorcycle / ATV Training: Personnel must complete required safety training before riding a motorcycle, and local units should address required PPE, licensing, refresher training, and safety representative contacts.

Pedestrian / Bicycle Awareness: Emphasize local roadway crossing hazards, visibility requirements, spotter use, and other pedestrian or bicycle precautions specific to the installation.

Local Traffic Hazards / Vehicle Notes:
Enter local gate hazards, base-specific traffic concerns, spotter rules, and where vehicle training is documented.

Motorcycle Safety Representatives / Traffic Contacts:
List unit or group motorcycle safety representatives, traffic safety contacts, and local rider briefing requirements.



No routine-use GVO risk assessment files uploaded.

Reporting Unsafe Equipment, Conditions, or Procedures

Notify the supervisor immediately when unsafe equipment, conditions, or procedures are identified. Personnel may report hazards without fear of retaliation, coercion, discrimination, or reprisal.

- **DAF Form 457:** Use to document and elevate unsafe conditions, equipment, or processes.
- **SAFEREP:** Use the approved mishap or hazard reporting channel for additional reporting when directed by local procedures.

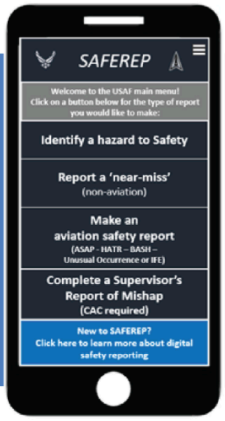
Safety Reporting … All In One Place

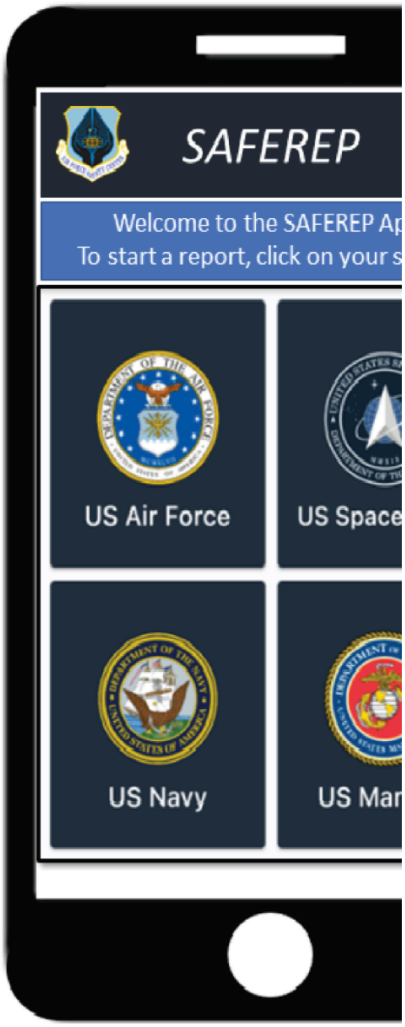
SAFEREP APP!

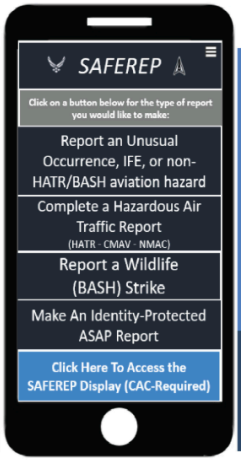
Why Safety Reporting?

Quickly and easily tell safety about safety issues you see

- In the dorms
- At work
- Anywhere on base
- In flight
- While maintaining aircraft
- On the ramp







What to report?

- Dangerous conditions
- "Near miss" events
- Aviation specific reporting
- Cultural issues that could lead to an accident, damage, or injury
- Supervisor mishap report

Report anonymously
Reports go direct to the safety office

New name, new website, new app, and a whole new look

SAFEREP is a web- and app-based way for members from every career field to identify hazards and report near-misses and even some mishaps directly to their safety offices using any computer or mobile device.

SAFEREP

Download to your mobile device from your preferred App Store

Or

Go to https://saferep.safety.af.mil









Documentation and Review

Document initial, refresher, and task-specific training in the work center record, review annually or when work conditions change, and maintain records per local records management requirements.

Bioenvironmental Workplace Survey: No Bioenvironmental Workplace Survey uploaded.

Applicable Job Hazard Analyses: No Job Hazard Analysis files uploaded.

Annual Review Log:
Record the supervisor annual review history here.

Trainee Name	Employee ID / Last 4	Date Trained	Trainer / Supervisor Signature